Dot Blot Protocol

Materials: nanocellulose membrane, blank buffer (25mL 1x PBS buffer + 1.25g Blotting-Grade Blocker), PBST 5% buffer, HRP conjugate antibody (15mL PBS + 15uL antibody; could be reused within two weeks and kept in 4C), detection buffer (from Opti-4CN Substrate kit, 13.5mL miliQ + 1500uL diluent + 300uL substrate).

Procedure:

1. Load sample. Cut the membrane in proper size, put it in a plastic plate. Use p20 to load protein samples on the membrane, 2uL per dot, could increase volume if the solution is viscous. Include both positive and negative control.
2. Pour blank buffer into the plate to cover the membrane. Incubate on a shaker at room temperature for 1 hour.
3. Discard the blank buffer. Pour PBST 5% buffer into the plate to cover the membrane and shake it for 3 minutes, repeat this process twice.
4. Discard PBST buffer. Pour 1x PBS to cover the membrane and shake for 3 minutes.
5. Discard PBS buffer. Pour antibody solution to cover the membrane and shake for 1 hour.
6. Recycle antibody solution to falcon tube and put back to the fridge.
7. Add detection buffer to cover the membrane to visualize protein dots. Shake the plate for around 20 minutes until the dots are visualized.